

DDA3020 Machine Learning

Tutorial: Recurrent Neural Networks and Transformer

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Instructions

This exercise sheet covers the main ideas of sequential data modeling, recurrent neural networks (RNNs), and transformers. You should understand why sequential data requires models that preserve order information, how a basic RNN updates its hidden state over time, how RNNs are trained, and why gradient vanishing and exploding are important issues. You should also know the motivation of LSTM and GRU, the main limitations of RNNs, and the key ideas of transformers, including input embeddings, positional encodings, self-attention, cross-attention, and the advantages of transformers over RNNs.

Useful Formulas

1. Basic RNN update

A recurrent neural network updates its hidden state by

$$h_t = f_W(h_{t-1}, x_t).$$

A common form introduced in the lecture is

$$h_t = \tanh(W_{hh}h_{t-1} + W_{hx}x_t), \quad y_t = W_{hy}h_t.$$

2. RNN training objective

For a sequence of length T , the cost function is

$$E(\theta) = \frac{1}{T} \sum_{t=1}^T L(y_t, \hat{y}_t),$$

where $\theta = \{W, W'\}$ denotes the learnable parameters.

3. Attention projections

Given an embedding matrix

$$X \in \mathbb{R}^{n \times d},$$

and parameter matrices

$$W_Q, W_K, W_V \in \mathbb{R}^{d \times d_0},$$

the query, key, and value matrices are

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

4. Self-attention formula

The self-attention output is defined as

$$Z = \text{attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_0}}\right) V.$$

The attention map is

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_0}}\right) \in \mathbb{R}^{n \times n}.$$

Lecture Exercises

Q1. Discuss the connection and difference between a recurrent neural network (RNN) and a feed-forward neural network. (*Source: p. 17*)

Q2. Consider the language modeling example with training sequence "hello" and vocabulary $[h, e, l, o]$.

- Explain the meaning of the statement: given characters $1, 2, \dots, t$, the model predicts character $t + 1$.
- Explain why one-hot vectors can be used to represent the input characters in this example.

(Source: p. 22)

Q3. In the language modeling example, suppose the current input is "h". What should the model predict as the next character? (Source: p. 23)

Q4. In the language modeling example, suppose the current input is "he". What should the model predict as the next character? (Source: p. 24)

Q5. In the language modeling example, suppose the current input is "hel". What should the model predict as the next character? (Source: p. 25)

Additional Exercises

Q6. Which of the following is an example of a many-to-one task?

- (A) Image classification
- (B) Image captioning
- (C) Text classification
- (D) Machine translation

Q7. Why are recurrent neural networks more suitable than ordinary feed-forward neural networks for sequential data?

Q8. Consider the RNN update

$$h_t = f_W(h_{t-1}, x_t).$$

Explain the meaning of each term in this expression.

Q9. Which statement is correct about an RNN?

- (A) It uses different parameters at every time step
- (B) It shares the same parameters across time steps
- (C) It ignores the previous hidden state
- (D) It cannot process variable-length sequences

Q10. The lecture defines the cost function

$$E(\theta) = \frac{1}{T} \sum_{t=1}^T L(y_t, \hat{y}_t).$$

Explain the meaning of y_t , $\hat{y}_t$, $L(\cdot, \cdot)$, and T .

Q11. What is the idea of truncated backpropagation through time?

Q12. Which of the following best describes gradient exploding?

- (A) Gradients become exactly zero at every time step
- (B) Gradients become extremely small during training
- (C) Gradients become exceptionally large, causing numerical instability
- (D) The model stops updating its parameters completely

Q13. What is gradient clipping, and how can it help during training?

Q14. What problem was LSTM designed to address?

Q15. Which statement about LSTM is correct?

- (A) LSTM guarantees that gradients will never vanish or explode
- (B) LSTM is identical to a basic RNN
- (C) LSTM helps with long-distance dependencies but does not completely guarantee the absence of vanishing or exploding gradients
- (D) LSTM cannot be trained with gradient-based methods

Q16. What is a gated recurrent unit (GRU), and how is it structurally different from LSTM?

Q17. State two major limitations of basic RNNs and their variants.

Q18. Why is parallelization difficult for RNNs?

- (A) Because RNNs process data sequentially across time steps
- (B) Because RNNs do not use matrix multiplication
- (C) Because RNNs have no hidden states
- (D) Because RNNs cannot be trained on GPUs

Q19. What is a transformer, and what mechanism is it based on?

Q20. Why are positional encodings needed in a transformer?

- (A) Because transformers process tokens in parallel and need extra information to represent order
- (B) Because positional encodings replace input embeddings
- (C) Because positional encodings reduce the vocabulary size
- (D) Because positional encodings are only used in CNNs

Q21. Explain the meanings of the matrices Q , K , and V in the attention mechanism.

Q22. What is the difference between self-attention and cross-attention?

Q23. Compare RNNs and transformers in terms of parallelization, handling long-range dependencies, and model capacity and flexibility.

Solutions

Q1. Answer: An RNN is related to a feed-forward neural network in that both are parameterized neural models trained by backpropagation, but an RNN is specifically designed for sequential data by introducing a recurrent hidden state and sharing parameters over time.

Explanation: Both RNNs and feed-forward neural networks are neural network models with learnable parameters. Both transform an input into a hidden representation and then produce an output, and both can be trained by backpropagation and gradient-based optimization. The difference is that a feed-forward neural network processes inputs independently and does not explicitly model temporal dependence, while an RNN is designed for sequential data and maintains a hidden state that is passed from one time step to the next. In an RNN, the same parameters are shared across time steps, so it can handle variable-length sequences and capture order information.

Q2. Answer: The model predicts the next character from the previous context, and one-hot vectors provide a simple way to represent each character as a discrete symbol from the vocabulary.

Explanation: The statement means that the model uses the characters seen so far as context and predicts the next character in the sequence. For example, in the word "hello", given "h", the model predicts "e"; given "he", it predicts "l"; and given

"he1", it predicts "1". One-hot vectors can be used because the vocabulary is finite and small. Each character corresponds to one unique position in the vocabulary, so it can be represented by a vector with one entry equal to 1 and all other entries equal to 0. For vocabulary $[h, e, l, o]$, the encoding can be $h = [1, 0, 0, 0]^T$, $e = [0, 1, 0, 0]^T$, $l = [0, 0, 1, 0]^T$, and $o = [0, 0, 0, 1]^T$.

Q3. Answer: "e"

Explanation: In the training sequence "hello", the character that follows "h" is "e".

Q4. Answer: "1"

Explanation: In the training sequence "hello", the character that follows the prefix "he" is "l".

Q5. Answer: "1"

Explanation: In the training sequence "hello", the character that follows the prefix "hel" is also "l".

Q6. Answer: C

Explanation: A many-to-one task takes a sequence as input and produces a single output. Text classification is a typical example because the model reads a sequence of words and outputs one label.

Q7. Answer: RNNs are better suited for sequential data because they maintain a recurrent hidden state, share parameters across time, and can model order and dependencies in variable-length sequences.

Explanation: Recurrent neural networks explicitly model temporal dependence. The hidden state passed from one time step to the next allows information from earlier elements in the sequence to influence later outputs. In contrast, ordinary feed-forward neural networks process inputs independently and do not naturally represent sequential structure.

Q8. Answer: h_t is the hidden state at time step t , h_{t-1} is the hidden state from the previous time step, x_t is the input at time step t , and f_W is the transformation function parameterized by W that combines the previous hidden state and the current input to produce the new hidden state.

Explanation: This recurrence is the core mechanism of an RNN. It updates the internal state step by step so that past information can affect future computation.

Q9. Answer: B

Explanation: An RNN uses the same parameters across all time steps.

Q10. Answer: y_t is the ground-truth target at time step t , $\hat{y}_t$ is the model prediction at time step t , $L(y_t, \hat{y}_t)$ is the loss at time step t , and T is the total number of time steps in the sequence.

Explanation: The total cost is the average loss over the whole sequence.

Q11. Answer: Truncated backpropagation through time carries hidden states forward through the whole sequence but only backpropagates gradients for a limited number of time steps.

Explanation: This approximates full backpropagation through time while reducing computational cost and memory usage.

Q12. Answer: C

Explanation: Gradient exploding means that gradients become exceptionally large during training, which can cause numerical instability.

Q13. Answer: Gradient clipping is a technique that restricts gradients to lie within a specified range or below a certain norm threshold. It helps prevent gradients from becoming excessively large and improves numerical stability during training.

Explanation: It is a practical remedy for gradient exploding.

Q14. Answer: LSTM was designed to address the vanishing gradient problem and to improve the modeling of long-distance dependencies.

Explanation: Basic RNNs struggle to propagate useful learning signals over long sequences, while LSTM provides an easier way for the model to learn long-range relationships.

Q15. Answer: C

Explanation: According to the lecture, LSTM helps with long-distance dependencies, but it does not completely guarantee the absence of vanishing or exploding gradients.

Q16. Answer: A gated recurrent unit (GRU) is a type of recurrent neural network that uses gating mechanisms to control the flow of information. Compared with LSTM, GRU has fewer parameters because it lacks an output gate.

Explanation: The lecture describes GRU as similar to LSTM but structurally simpler.

Q17. Answer: Two major limitations are:

- difficulty in capturing long-term dependencies because of the vanishing gradient problem,
- limited parallelization because RNNs process data sequentially.

Explanation: These two limitations motivate the use of more advanced variants and transformer-based models.

Q18. Answer: A

Explanation: RNN computation at each time step depends on the previous time step, so operations cannot be fully parallelized across the sequence.

Q19. Answer: A transformer is a deep learning architecture based on the attention mechanism, especially multi-head attention.

Explanation: Unlike RNNs, transformers rely on attention rather than recurrence to model relationships in sequences.

Q20. Answer: A

Explanation: Transformers process tokens in parallel, so they need positional encodings to inject order information into the model.

Q21. Answer: Q , K , and V are the query, key, and value matrices. They are obtained by linear transformations of the input embeddings:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Explanation: In attention, queries are matched against keys to determine importance weights, and values are then combined according to those weights.

Q22. Answer: In self-attention, Q , K , and V are computed from the same feature matrix. In cross-attention, they are computed from different features.

Explanation: Self-attention models relationships within one sequence, while cross-attention models relationships between different sequences or feature sources.

Q23. Answer: Compared with RNNs, transformers are easier to parallelize, more effective in handling long-range dependencies, and usually have higher model capacity and flexibility.

Explanation: RNNs process sequences step by step and often struggle with long-term dependencies, while transformers use attention to connect tokens more directly and can compute over sequences in parallel.